PValue

Data Science Meetup Pavia

Time Series Analysis and Forecasting

Forecasting gas demand: a Machine Learning approach

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1. The data science process

A data science workflow

6. DEPLOYMENT

- Architecture design
- Execution and re-train scheduling
- Industrialization

5. EVALUATION

- Performance evaluation and testing

4. MODELING

- Modeling
- Performance evaluation

1. BUSINESS UNDERSTANDING

- Problem statement
- Evaluation metrics

2. DATA UNDERSTANDING

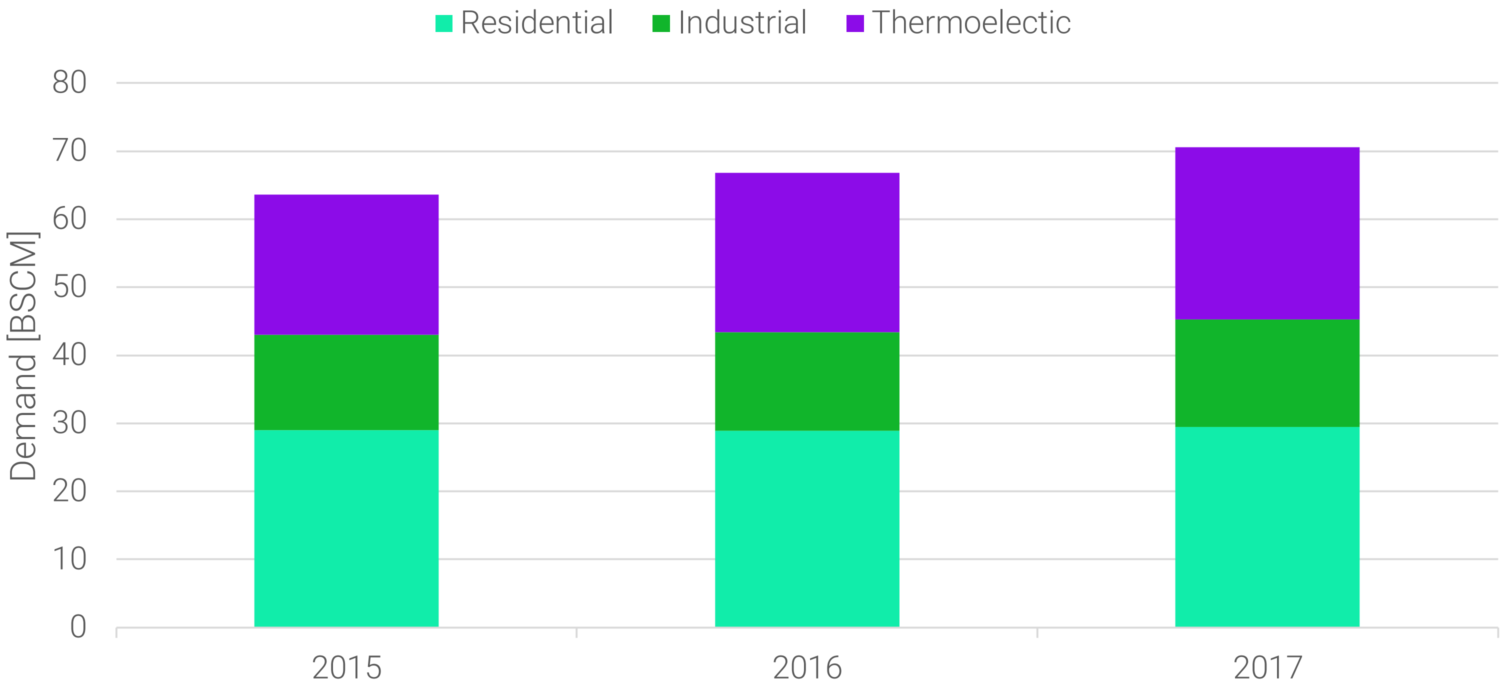
- Source discovery
- Quality assessment
- Data exploration

3. DATA PREPARATION

- Data cleaning
- Feature engineering

2. Business Understanding

Context



Source: SNAM Rete Gas, http://pianodecennale.snamretegas.it/it/domanda-offerta-di-gas-in-italia/domanda-di-gas-naturale.html

Motivation

Pipe reservation

Energy companies need to reserve pipe capacity in **advance**. Accurate demand models can prevent **inefficiency** in reservations.

Network balance

The Transmission System Operator (TSO) applies financial **penalties** in case of network unbalance. Accurate demand forecasting decreases risk of unbalance.

Price forecasting

Demand is one of the main inputs to gas **price** models, which are key to design working schedules for **power plants** and other strategic business decisions

Problem statement

Given:

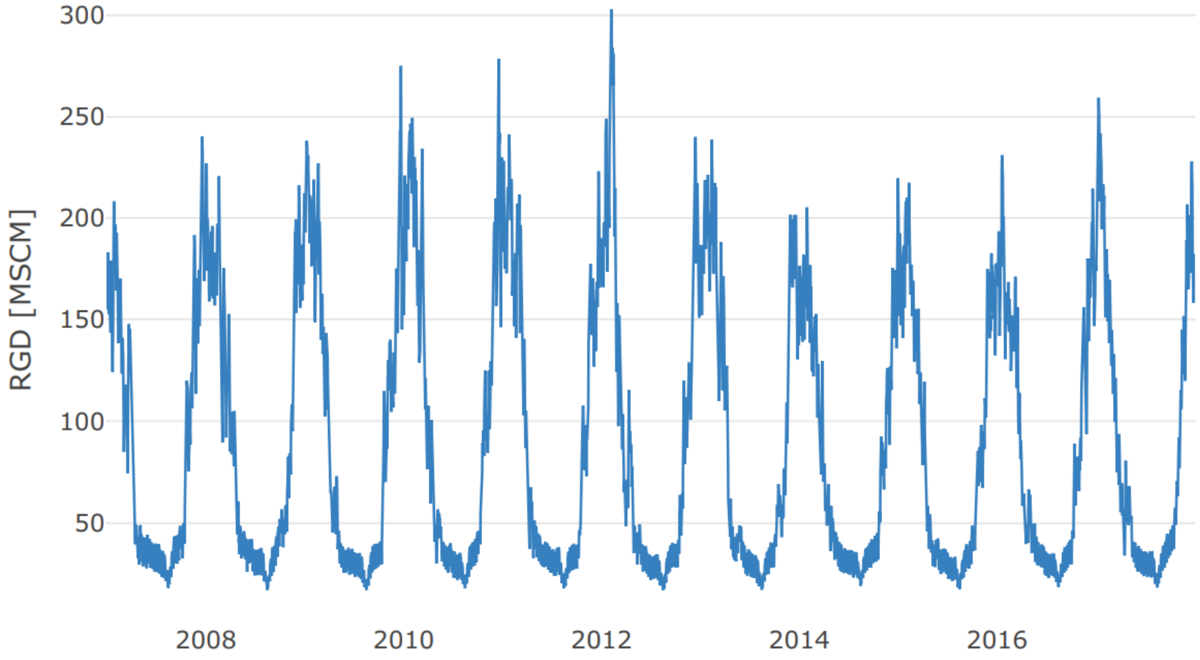
- Daily Italian residential, industrial and thermoelectric demand from 2007 to 2018
- Forecasts of average daily temperature for Northern Italy from 2007 to 2018
- Actual average daily temperature for Northern Italy from 2015 to 2018

Perform:

- Day-ahead prediction of residential, industrial and thermoelectric gas demand
- Day-ahead prediction of Italian gas demand

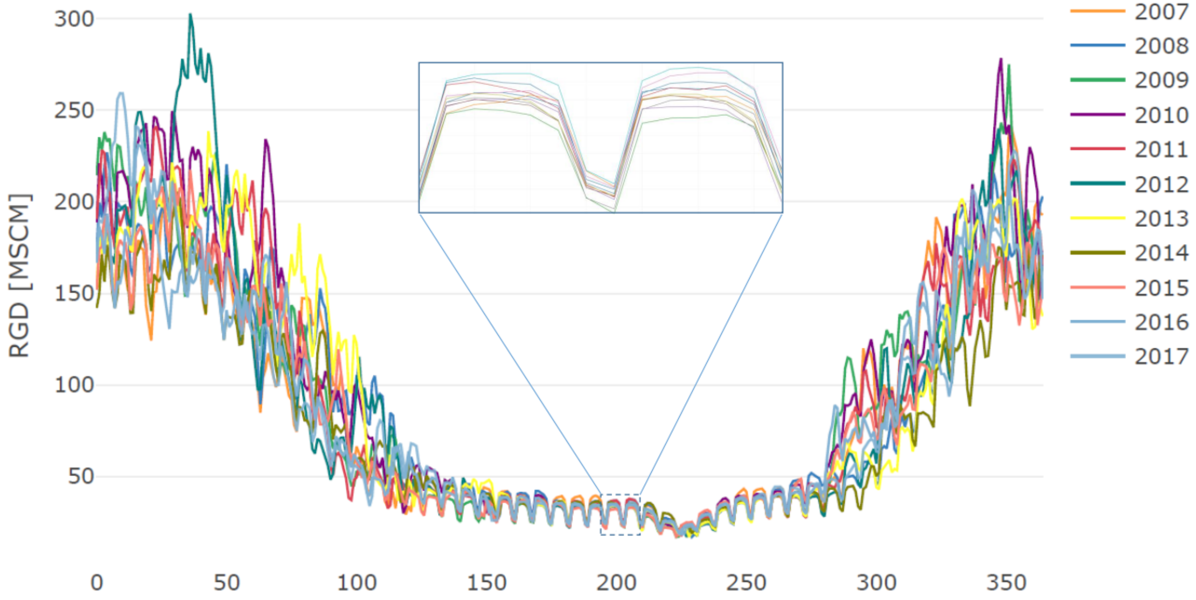
3. Data Understanding

Residential demand



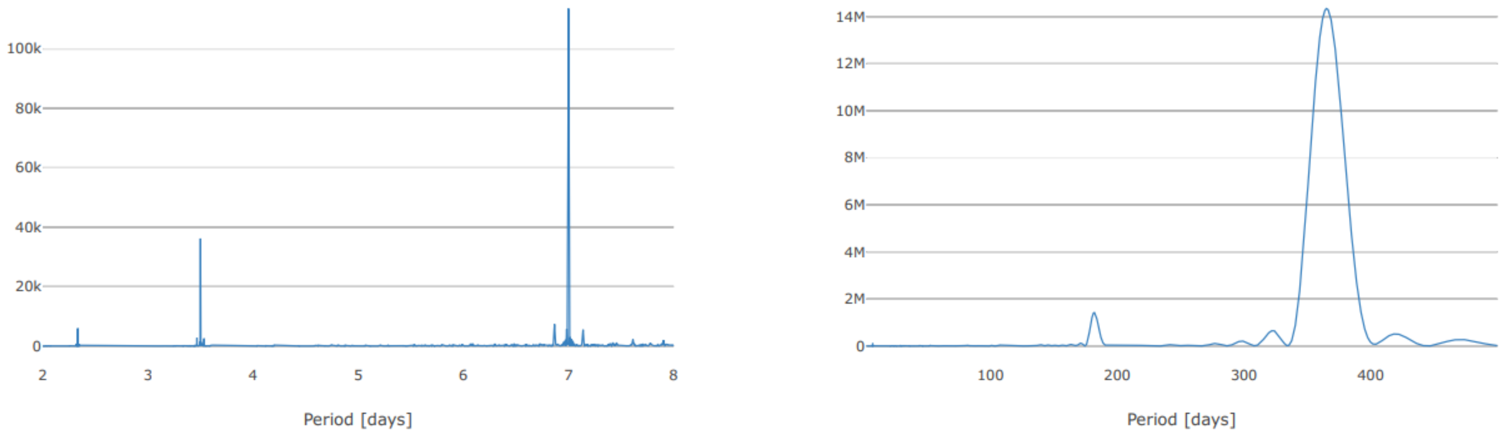
Italian daily residential gas demand (RGD).

Residential demand
Seasonality



Italian daily residential gas demand (RGD). Time series are shifted to align weekdays: weekly periodicity is particularly visible in summer. The yearly seasonal variation is mostly explained by heating requirements. In the inset, two weeks of July's demand data are zoomed

Residential demand
Seasonality



Periodogram of Italian daily residential gas demand (RGD). Left panel: periods from 0 to 8 days; right panel: periods from 0 to 500 days. The yearly periodicity is highlighted by peaks at 365.25 days, while the weekly one by the smaller spike at a period of 7 days. Other notable values are caused by harmonics

Residential demand
Similar day

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Definition 1 (Similar Day). *If t is not a holiday, its similar day* $\tau^* = \mathrm{sim(t)}$ *is*

$$\tau^* = \arg\min_{\tau} |\mathrm{yearday}(\tau) - \mathrm{yearday(t)}|$$

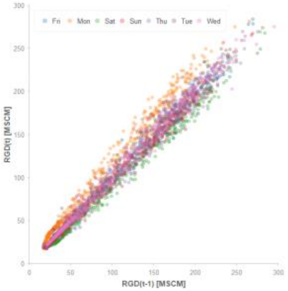
subject to

- $\mathrm{year}(\tau) = \mathrm{year(t)} - 1$;
- $\mathrm{weekday}(\tau) = \mathrm{weekday(t)}$;
- τ *is not a holiday.*

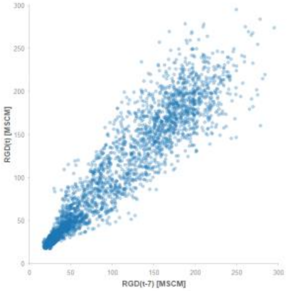
If t is a holiday, its similar day $\tau^* = \mathrm{sim(t)}$ *is the same holiday in the previous year.*

According to the Italian calendar, holidays are: 1 January, 6 January, 25 April, 1 May, 2 June, 15 August, 1 November, 8, 25 and 26 December, Easter and Easter Monday.

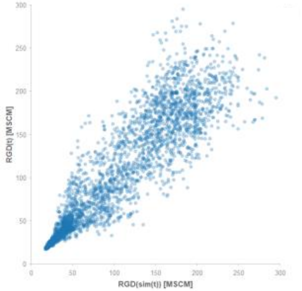
Residential demand
Features



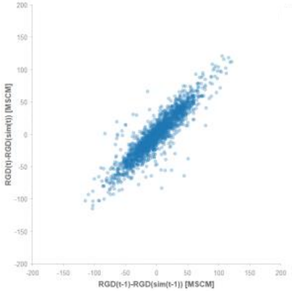
(a) $RGD(t)$ vs $RGD(t-1)$



(b) $RGD(t)$ vs $RGD(t-7)$



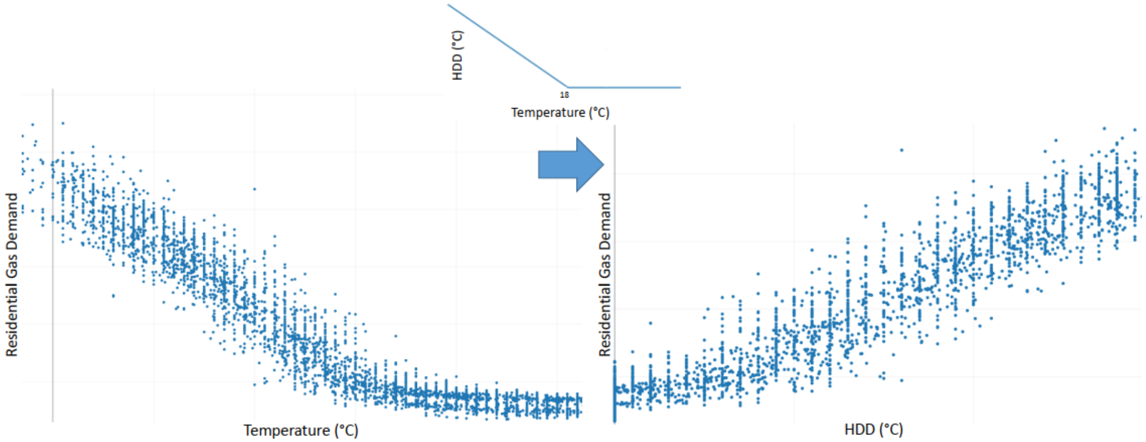
(c) $RGD(t)$ vs $RGD(\mathrm{sim(t)})$



(d) $RGD(t) - RGD(\mathrm{sim(t)})$ vs $RGD(t-1) - RGD(\mathrm{sim(t-1)})$

Scatter plots between residential gas demand (RGD) and potential features to be used for its prediction. It is possible to note that RGD at times t-1, t-7 and sim(t) is highly correlated with RGD at time t. Moreover, RGD(t-1)-RGD(sim(t-1)) appears to be a good proxy of RGD(t) – RGD(sim(t))

Residential demand
Temperature



Relation between Italian daily residential gas demand (RGD) and temperature.
Left panel: scatter plot of daily RGD vs average daily temperature. Right panel: scatter plot of daily RGD vs HDD. Inset: HDD as a function of the temperature.
The relation between HDD and RGD is approximately linear

Residential demand Temperature

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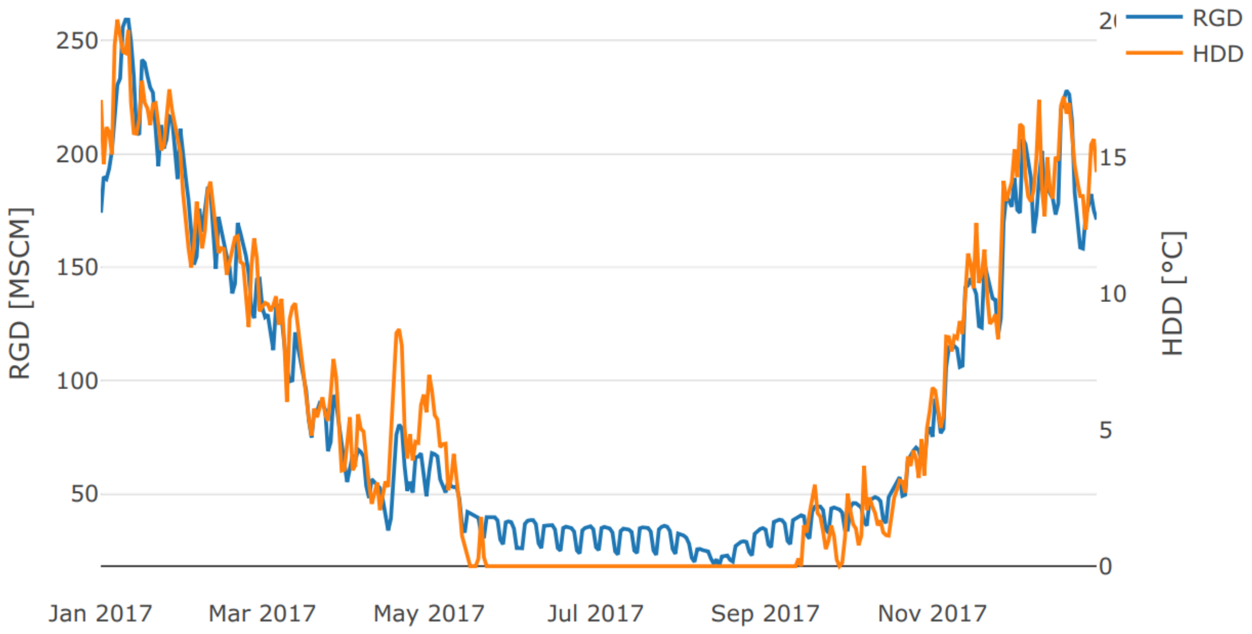
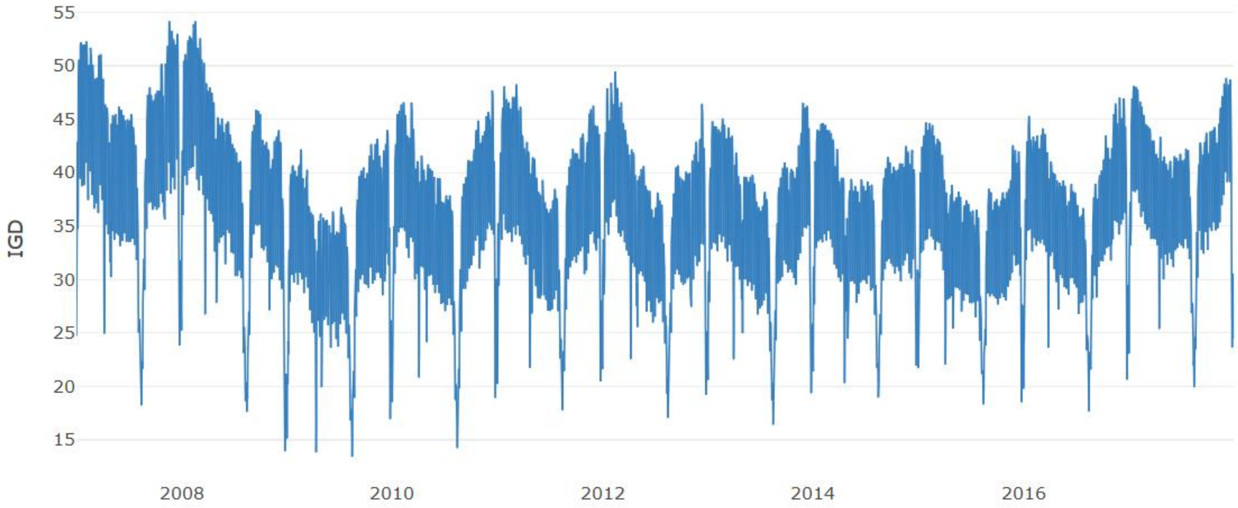


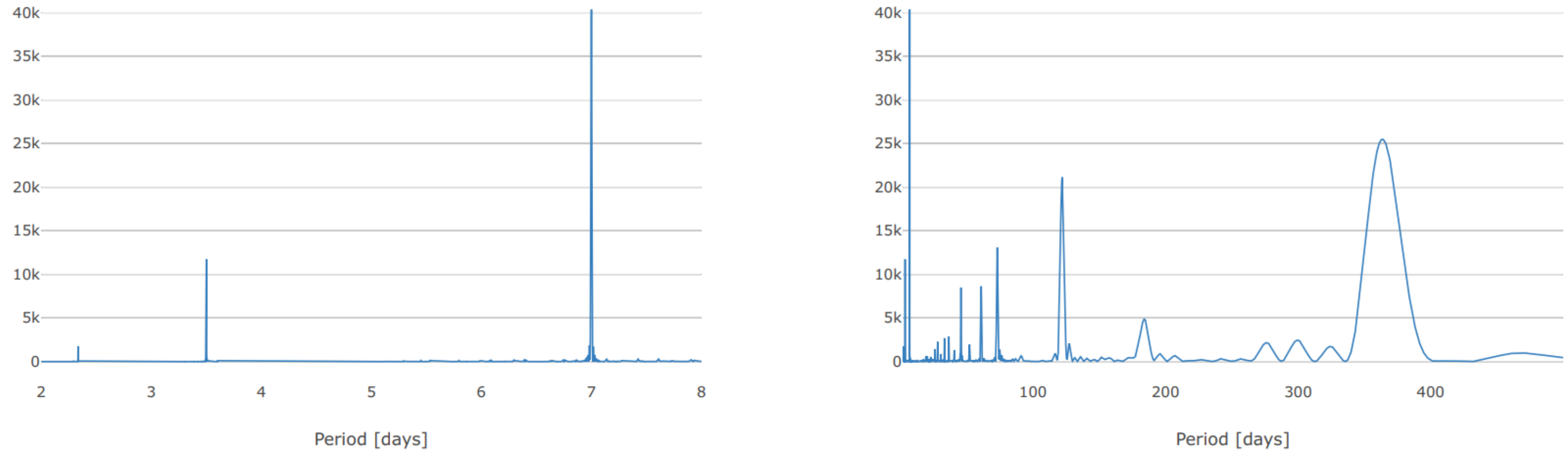
Time series of residential gas demand (RGD) and Heating Day Degrees (HDD) in 2017.
The instantaneous correlation between the two series is apparent

Industrial demand



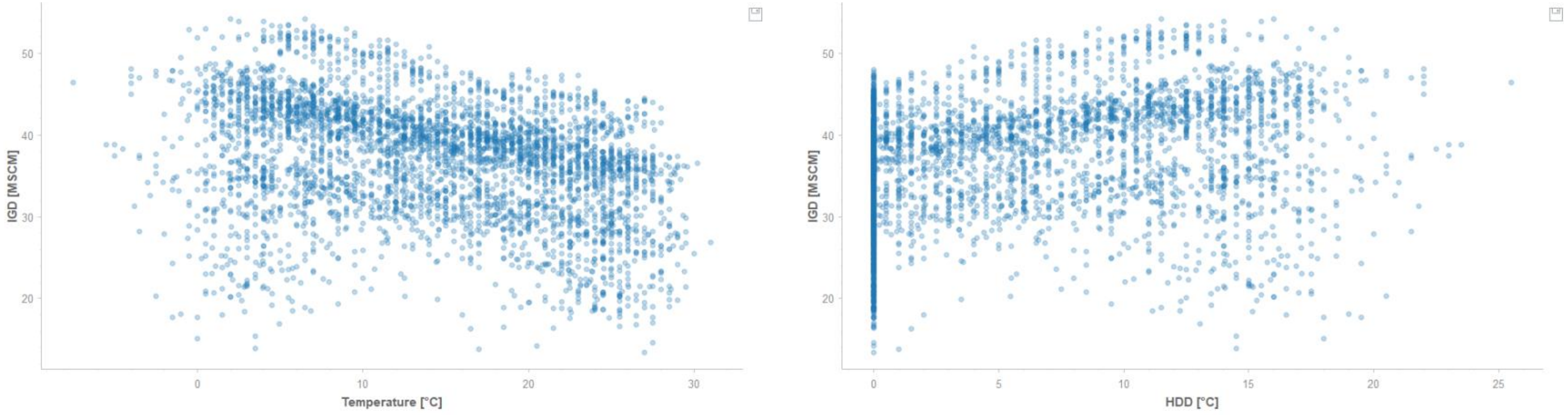
Italian daily industrial gas demand (IGD). The decrease in average value in 2009 is an effect of the economic crisis started in 2008, while negative peaks are Christmas, Easter and summer holidays

Industrial demand
Seasonality



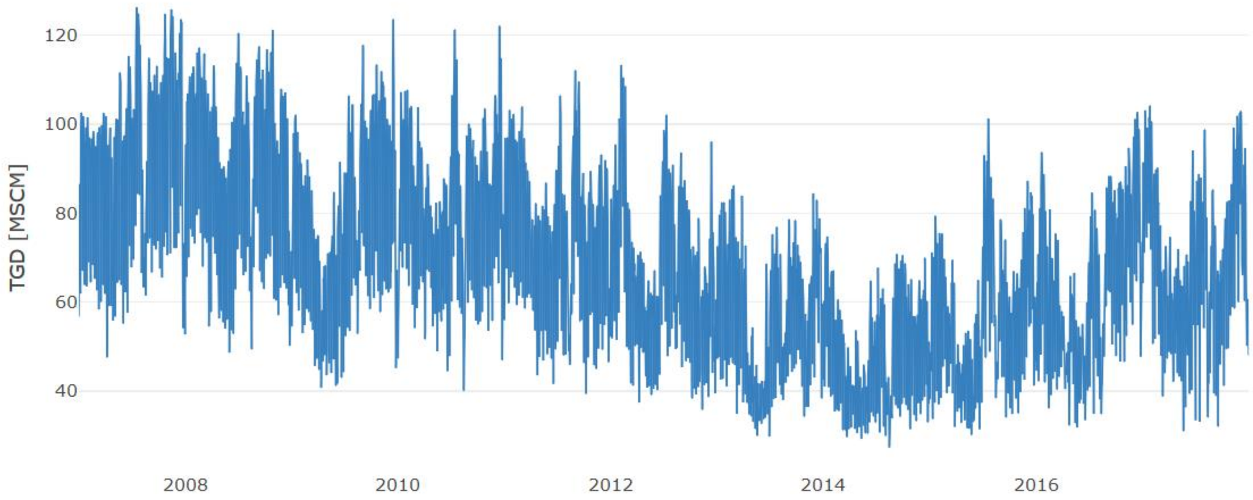
Periodogram of Italian daily industrial gas demand (IGD). Left panel: periods from 0 to 8 days; right panel: periods from 0 to 500 days. Notably, the peak ascribable to weekly periodicity is here higher than the one produced by yearly seasonality

Industrial demand
Temperature



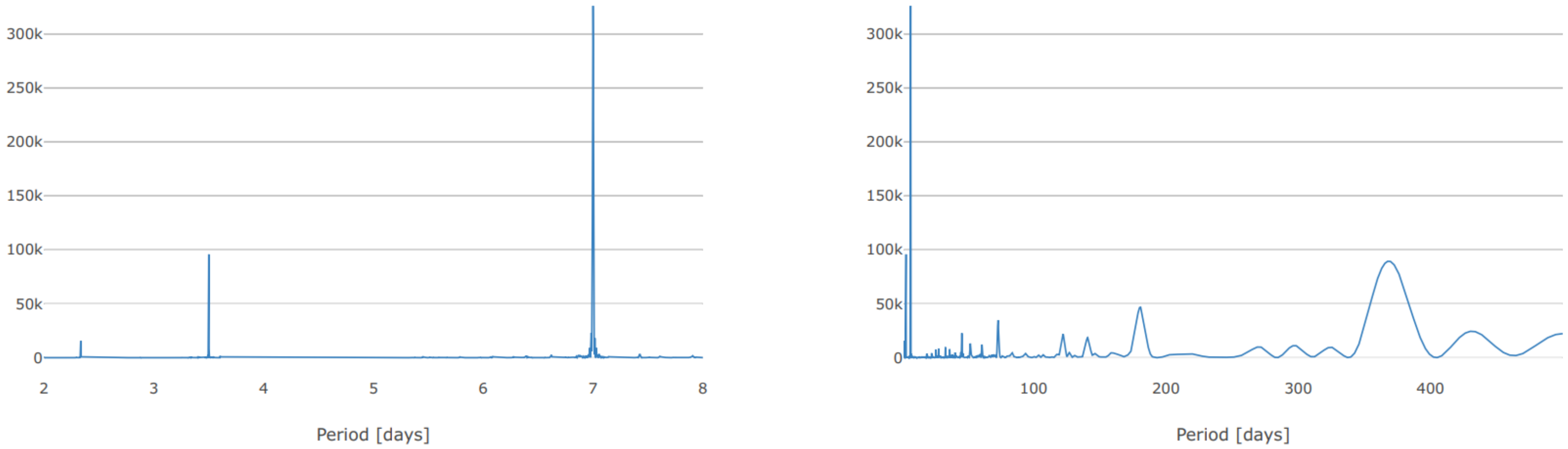
Relation between Italian daily industrial gas demand (IGD) and temperature.
Left panel: scatter plot of daily IGD vs average daily temperature. Right panel: scatter plot of daily IGD vs HDD. The relation between IGD and temperature looks linear in the whole range of values, thus the introduction of HDD does not increase the correlation

Thermoelectric demand



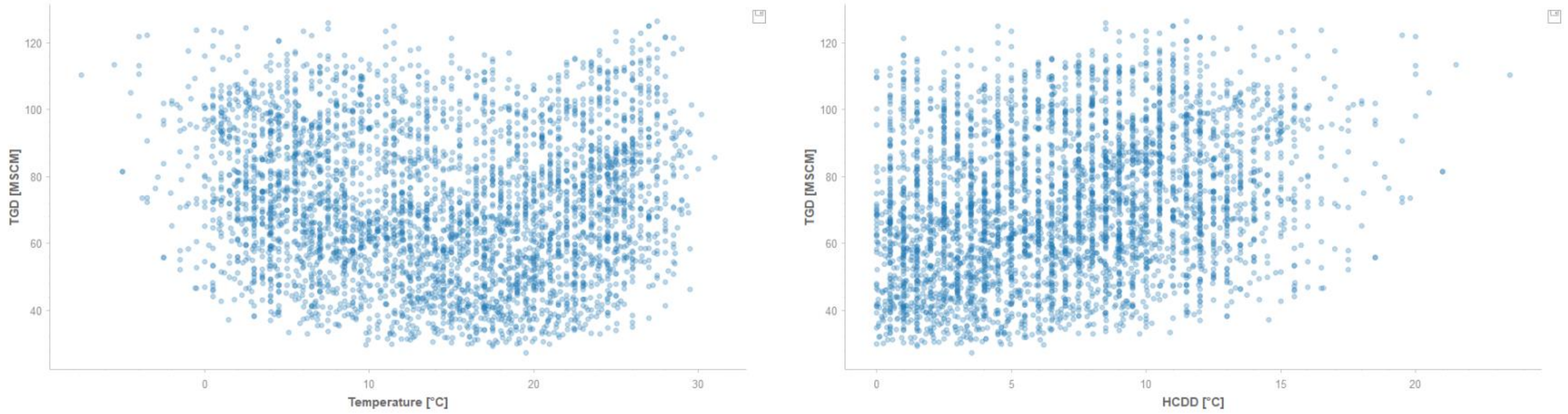
Italian daily thermoelectric gas demand (TGD).
The decreasing trend from 2010 to 2014 is explained by the rise of renewable power sources, which slowed down since 2015

Thermoelectric demand
Seasonality



Periodogram of Italian daily thermoelectric gas demand (TGD).
Left panel: periods from 0 to 8 days; right panel: periods from 0 to 500 days. Due to several exogenous factors which affectTGD (like power price and gas price), yearly periodicity is less important than in IGD and RGD

Industrial demand
Temperature



Relation between Italian daily thermoelectric gas demand (TGD) and temperature.
Left panel: scatter plot of daily TGD vs average daily temperature. Right panel: scatter plot of daily TGD vs Heating and Cooling Day Degrees (HCDD), defined as HCDD = |temperature -16°C|. The influence of temperature on TGD is similar to the one on power demand

Exploratory analysis
Power load

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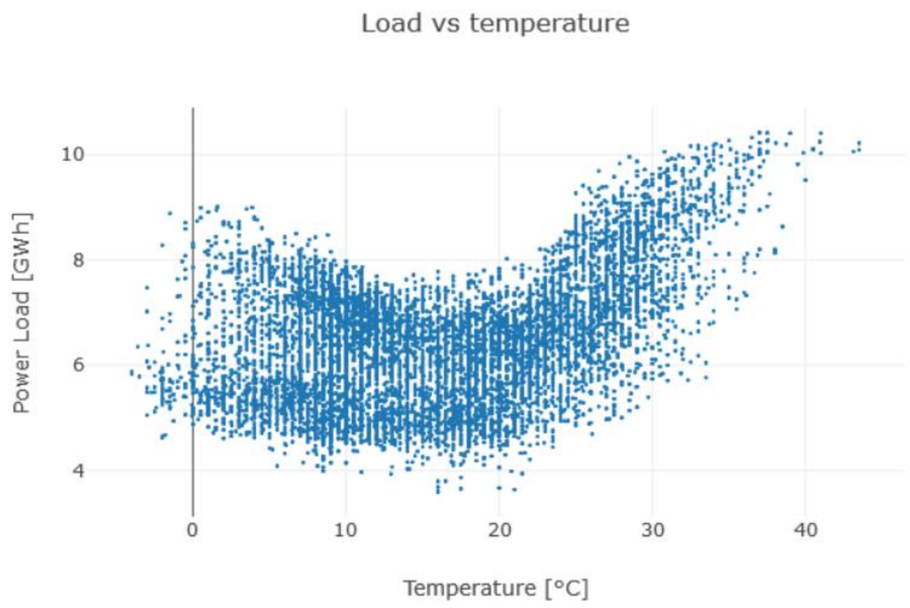

Relation between Greek hourly power load and temperature.
The shape of the scatter plot is very similar to the one representing TGD and temperature.
Source: https://towardsdatascience.com/lessons-from-a-real-machine-learning-project-part-2-the-traps-of-data-exploration-e0061ace84aa

4. Data Preparation

Final features

Feature	Reference time	Type	Series
Gas demand series	t-1	continuous	RGD, IGD, TGD
Gas demand series	t-7	continuous	RGD, IGD, TGD
Gas demand series	$\mathrm{sim}(t)$	continuous	RGD, IGD, TGD
Gas demand series	$\mathrm{sim}(t-1)$	continuous	RGD, IGD, TGD
Forecasted temperature	t	continuous	RGD, IGD, TGD
Forecasted temperature	t-1	continuous	RGD, IGD, TGD
Forecasted temperature	t-7	continuous	RGD, IGD, TGD
Forecasted temperature	$\mathrm{sim}(t)$	continuous	RGD, IGD, TGD
Forecasted HDD	t	continuous	RGD, IGD
Forecasted HDD	t-1	continuous	RGD, IGD
Forecasted HDD	t-7	continuous	RGD, IGD
Forecasted HDD	$\mathrm{sim}(t)$	continuous	RGD, IGD
Forecasted HCDD	t	continuous	TGD
Forecasted HCDD	t-1	continuous	TGD
Forecasted HCDD	t-7	continuous	TGD
Forecasted HCDD	$\mathrm{sim}(t)$	continuous	TGD
Weekday	t	categorical	RGD, IGD, TGD
Holiday	t	dummy	RGD, IGD, TGD
Day after holiday	t	dummy	RGD, IGD, TGD
Bridge holiday	t	dummy	RGD, IGD, TGD

5. Modeling

Models

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Basic models

- **Regularized linear models**: lasso, ridge, elastic net
- **Non-linear models**: Torus model, support vector regression, random forest, fully-connected neural network
- **Non-parametric models**: Gaussian process, nearest neighbours

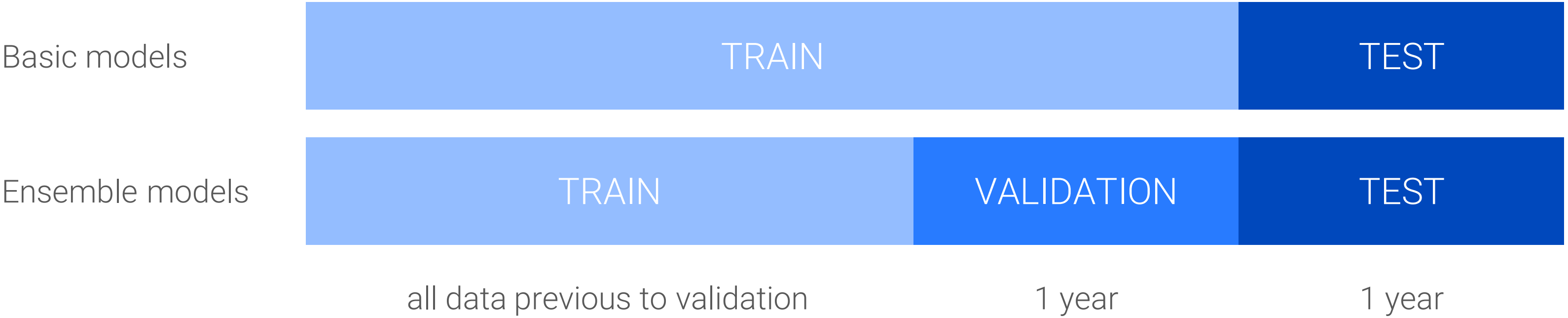
Ensemble models

- Simple average
- Weighted average
- Subset average
- Support vector regression

6. Evaluation

Experiments

- Five one-year-long test sets, from 2014 to 2018, to assess out-of-sample performance
- To each test set is associated a validation set, covering the previous year, to train ensemble models
- All the data previous to the start of the validation set are included in train set of basic models
- Standalone basic models (i.e. basic models not used for ensembling) are trained on the union of train and validation sets
- Mean Absolute Error (MAE) is used as a performance metric



Residential demand
Results

Model	2015	2016	2017	2018	Average
Ridge	3.39	3.10	3.01	3.49	3.25
Lasso	3.38	3.10	3.01	3.49	3.25
Elastic net	3.38	3.10	3.01	3.49	3.25
SVR	2.84	2.62	2.38	2.93	2.69
GP	2.60	2.48	2.51	2.61	2.55
KNN	4.57	5.51	5.08	5.52	5.17
Random forest	3.04	3.36	3.50	3.48	3.35
Torus	3.18	2.66	2.54	3.13	2.88
ANN	2.76	2.68	2.43	3.10	2.74
Simple average	2.66	2.57	2.45	2.91	2.65
Subset average	**2.41**	**2.17**	**2.06**	**2.56**	**2.30**
Weighted average	2.59	2.33	**2.06**	2.64	2.40
SVR aggregation	2.58	2.30	2.19	2.67	2.44

Out-of-sample MAEs for Residential Gas Demand

Industrial demand
Results

Model	2015	2016	2017	2018	Average
Ridge	0.75	0.75	0.74	0.77	0.75
Lasso	0.75	0.75	0.74	0.77	0.75
Elastic Net	0.75	0.75	0.74	0.77	0.75
SVR	0.57	0.58	0.7	0.75	0.65
GP	0.61	0.61	0.68	0.70	0.65
KNN	1.46	1.25	1.95	1.23	1.47
Random Forest	0.78	0.86	0.95	0.83	0.86
Torus	0.96	0.97	1.05	1.10	1.02
ANN	0.66	0.80	**0.57**	0.74	0.69
Simple average	0.60	0.62	0.69	0.66	0.64
Subset average	0.56	0.56	0.58	**0.61**	**0.58**
Weighted average	**0.55**	**0.55**	0.65	0.70	0.61
SVR aggregation	0.57	0.79	**0.57**	0.81	0.68

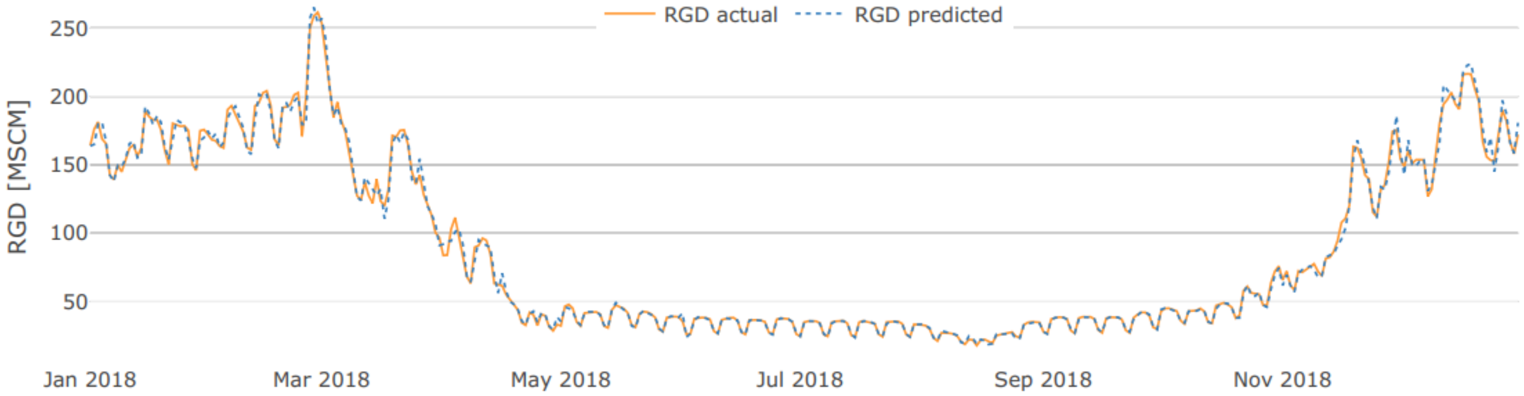
Out-of-sample MAEs for Industrial Gas Demand

Thermoelectric demand Results

Model	2015	2016	2017	2018	Average
Ridge	3.73	4.15	4.26	4.48	4.15
Lasso	3.73	4.15	4.26	4.49	4.16
Elastic Net	3.73	4.15	4.26	4.49	4.16
SVR	3.41	3.64	4.33	4.33	3.93
GP	3.49	3.70	4.39	4.34	3.98
KNN	6.13	5.22	5.83	5.54	5.68
Random Forest	4.66	4.43	4.87	4.84	4.70
Torus	3.98	4.48	4.96	4.94	4.59
ANN	3.40	3.97	4.32	4.41	4.03
Simple average	3.50	3.75	4.21	4.36	3.96
Subset average	**3.26**	3.65	**4.17**	**4.26**	**3.84**
Weighted average	3.35	3.71	4.31	4.31	3.92
SVR aggregation	3.38	**3.62**	4.28	4.37	3.91

Out-of-sample MAEs for Thermoelectric Gas Demand

Residential demand
Actual vs forecast



Actual vs predicted Residential Gas Demand. Error is very close to zero during the summer and higher during the winter.

Thermoelectric demand
Actual vs forecast

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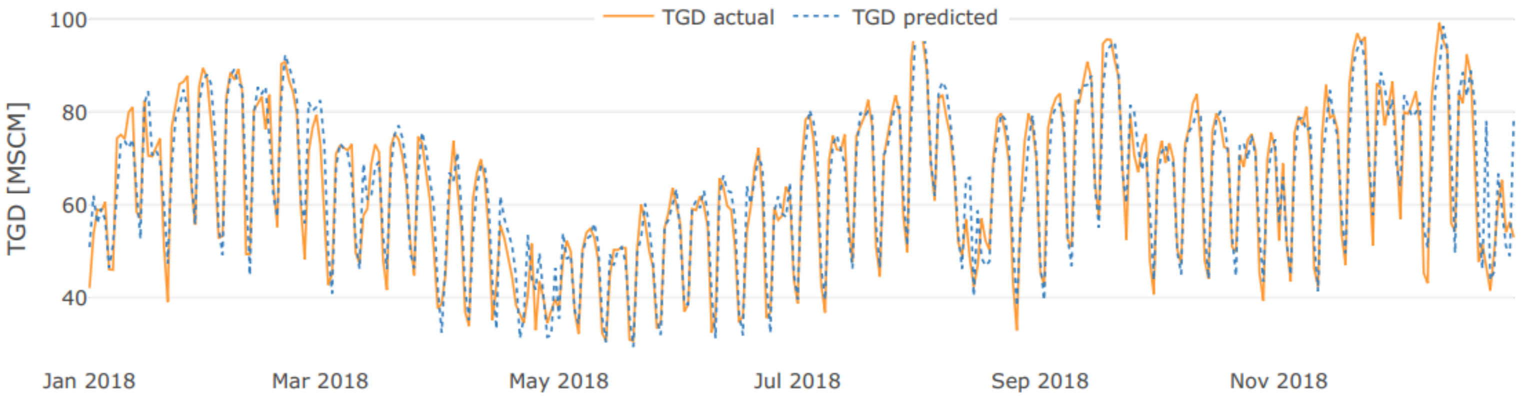


Actual vs predicted Thermoelectric Gas Demand. Error is higher with respect to residential demand due to the effect of multiple exogenous factors, such as power prices and renewable production.

Comparison with state-of-the-art

A comparison with SNAM, the Italian transmission system operator, is possible only for the overall Italian gas demand, intended as sum of residential, industrial and thermoelectric components Comparison was performed on 2017, the last year where SNAM forecasts are completely available.

5.16 MSCM

MAE achieved by our best ensemble model

9.57 MSCM

MAE achieved by SNAM, Italian transmission system operator

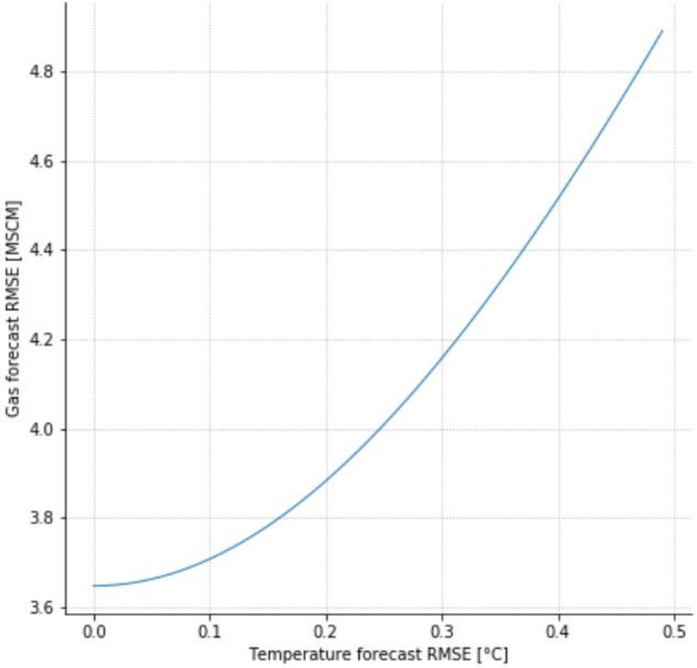
7. Deployment

From Jupyter to Luigi

More at: https://towardsdatascience.com/lessons-from-a-real-machine-learning-project-part-1-from-jupyter-to-luigi-bdfd0b050ca5

8. A side note

Effect of temperature forecasting error



Influence of temperature forecasting error on gas forecasting error.
When the temperature error is small, its contribution to the gas demand forecasting error is negligible, but the relationship between the temperature RMSE and gas demand RMSE becomes linear as the former increases.

9. Acknowledgements

Giuseppe De Nicolao,
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A2A

Life is about passion.
Thank you for sharing ours.

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Questions?